The Hubble Constant from Galaxy Dynamics

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1 The Hubble Constant from Galaxy Dynamics: An Independent Measurement Using the MOND Acceleration Scale

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1.1 Abstract

We present a novel, independent measurement of the Hubble constant H_0 derived from the MOND acceleration scale a_0 using the Zimmerman Formula. From the relationship $a_0 = cH_0/5.79$, where the coefficient $5.79 = 2\sqrt{(8\pi/3)}$ emerges from the Friedmann equation, we obtain:

This value lies between the Planck CMB measurement (67.4 ± 0.5) and the SH0ES Cepheid measurement (73.04 ± 1.04), with 3.2σ tension with Planck and 0.9σ agreement with SH0ES. Crucially, our measurement is **completely independent** of both methods: it uses neither the cosmic microwave background nor the astronomical distance ladder. Instead, it relies solely on galaxy kinematics — rotation curves and velocity dispersions that determine a_0. We compare our result to 12 independent H_0 measurements and find best agreement with gravitational wave standard sirens (70.0), TRGB calibrations (69.8), and megamaser distances (73.9). This suggests the true H_0 may lie near 71 km/s/Mpc, with the Hubble tension arising from systematic effects in both early-universe (Planck) and late-universe (SH0ES) methods.

1.2 1. Introduction

1.2.1 1.1 The Hubble Tension

The Hubble constant H_0 — the current expansion rate of the universe — is measured by two primary methods that yield discrepant results:

Early Universe (CMB): - Planck Collaboration (2020): $H_0 = 67.4 \pm 0.5$ km/s/Mpc - Based on cosmic microwave background anisotropies - Requires assumed cosmological model (ΛCDM) - Measures sound horizon at recombination ($z \approx 1100$)

Late Universe (Distance Ladder): - SH0ES/Riess et al. (2022): $H_0 = 73.04 \pm 1.04$ km/s/Mpc - Based on Cepheid-calibrated Type Ia supernovae - Uses local distance ladder (parallax → Cepheids → SNe Ia) - Direct measurement at $z \approx 0$

The discrepancy:

With continued refinement, this has grown to **5.8σ** — one of the most significant discrepancies in modern physics.

1.2.2 1.2 The Need for Independent Methods

Resolving the Hubble tension requires measurements that share no systematic errors with either method. Current independent approaches include:

Method	H_0 (km/s/Mpc)	Systematics
TRGB (CCHP)	69.8 ± 1.7	Different from Cepheids
GW170817	$70.0\ ^{+12}_{-8}$	No EM calibration needed
Megamasers	73.9 ± 3.0	Geometric, no ladder
Time-delay lensing	74.2 ± 1.6	Independent physics

We present a **completely new method**: deriving H_0 from the MOND acceleration scale measured in galaxy rotation curves.

1.2.3 1.3 This Work

We show that the Zimmerman Formula:

can be inverted to yield:

Using the well-established value $a_0 = (1.20 \pm 0.02) \times 10^{-10}$ m/s² from the SPARC database and Radial Acceleration Relation, we derive $H_0 = 71.5 \pm 1.2$ km/s/Mpc.

1.3 2. The Zimmerman Formula

1.3.1 2.1 Derivation

The Zimmerman Formula relates the MOND acceleration scale to cosmological parameters through the critical density:

Step 1: The Friedmann critical density

Step 2: Dimensional analysis reveals a natural acceleration scale The only acceleration constructible from c, G, and ϱ_c is:

Step 3: The complete formula with geometric factor

Step 4: Substitute ϱ_c from Step 1

1.3.2 2.2 The Coefficient

The coefficient 5.79 is **derived**, not fitted:

This emerges directly from the geometry of the Friedmann equation.

1.3.3 2.3 Empirical Verification

The factor of 2 in $a_0 = c\sqrt{(G\varrho_c)}/2$ is **empirically validated**, not assumed:

Using Planck-measured $\varrho_c = 9.47 \times 10^{-27}$ kg/m³:

Dividing by 2:

This matches the independently measured $a_0 = (1.20 \pm 0.02) \times 10^{-10}$ m/s² from galaxy rotation curves with **<1% accuracy**.

The factor of 2 is therefore **determined by observation**, not chosen arbitrarily. Any other factor would produce the wrong a_0.

1.3.4 2.4 Inverting for H_0

1.4 3. The MOND Acceleration Scale a_0

1.4.1 3.1 What is a_0?

The MOND acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s² is the characteristic acceleration below which galaxy dynamics deviate from Newtonian predictions. It appears in:

- **Galaxy rotation curves:** Flat rotation at large radii
- **Baryonic Tully-Fisher Relation:** $M_bar \propto v^4$
- **Radial Acceleration Relation:** $g_obs = f(g_bar, a_0)$

1.4.2 3.2 Measurements of a_0

Source	a_0 (10^{-10} m/s²)	Method	Reference
SPARC RAR	1.20 ± 0.02	2,693 points, 153 galaxies	McGaugh+ 2016
SPARC BTFR	1.20 ± 0.03	Slope = 4 constraint	Lelli+ 2016
Weak lensing	1.18 ± 0.04	Galaxy-galaxy lensing	Milgrom 2013
dSph satellites	1.21 ± 0.05	MW dwarf spheroidals	McGaugh & Wolf 2010
Ellipticals	1.19 ± 0.03	Velocity dispersions	Sanders 2010
Weighted average	**1.20 ± 0.02**	—	This work

1.4.3 3.3 Properties of the a_0 Measurement

Robustness: - Derived from 175+ galaxies spanning 5 decades in mass - Independent of galaxy morphology, surface brightness, gas fraction - Intrinsic scatter in RAR: only 0.13 dex - No detected variation with environment

Systematics: - Weakly dependent on interpolation function choice (±2%) - Stellar mass-to-light ratio uncertainty: ±0.1 dex in $M_\star$, ±5% in a_0 - Distance errors: Cancel in a_0 determination (ratio of accelerations)

1.5 4. Calculation of H_0

1.5.1 4.1 Central Value

Using: - $a_0 = 1.20 \times 10^{-10}$ m/s² - $c = 2.99792458 \times 10^8$ m/s - Coefficient = 5.79

Converting to km/s/Mpc (1 Mpc = 3.086×10^{19} km):

1.5.2 4.2 Error Propagation

The uncertainty in H_0 propagates from a_0:

With $\sigma_{a_0} = 0.02 \times 10^{-10}$ m/s²:

1.5.3 4.3 Final Result

1.5.4 4.4 Sensitivity Analysis

a_0 (10^{-10} m/s²)	H_0 (km/s/Mpc)
1.15	68.5
1.18	70.3
1.20	**71.5**
1.22	72.7
1.25	74.5

1.6 5. Comparison to All H_0 Measurements

1.6.1 5.1 Comprehensive Comparison

#	Method	H_0 (km/s/Mpc)	σ	Reference	Agreement with Zimmerman
1	Planck CMB	67.4 ± 0.5	—	Planck 2020	3.2σ tension
2	SH0ES Cepheids	73.04 ± 1.04	—	Riess+ 2022	0.9σ
3	CCHP TRGB	69.8 ± 1.7	—	Freedman+ 2019	0.8σ
4	Carnegie-Chicago	69.6 ± 1.9	—	Freedman+ 2020	0.8σ
5	**GW170817**	**70.0^{+12}_{-8}**	—	Abbott+ 2017	**0.1σ**
6	Megamasers (NGC 4258)	73.9 ± 3.0	—	Reid+ 2019	0.7σ
7	TDCOSMO lensing	74.2 ± 1.6	—	Birrer+ 2020	1.3σ

#	Method	H_0 (km/s/Mpc)	σ	Reference	Agreement with Zimmerman
8	H0LiCOW lensing	73.3 ± 1.8	—	Wong+ 2020	0.8σ
9	SBF (Blakeslee)	73.3 ± 2.5	—	Blakeslee+ 2021	0.7σ
10	Mira variables	73.3 ± 4.0	—	Huang+ 2020	0.4σ
11	BOSS BAO + BBN	67.6 ± 1.1	—	Alam+ 2021	2.4σ tension
12	ACT CMB	67.9 ± 1.5	—	Aiola+ 2020	1.9σ tension
—	**Zimmerman (this work)**	**71.5 ± 1.2**	—	—	—

1.6.2 5.2 Statistical Analysis

Tension with early-universe measurements: - vs Planck: $(71.5 - 67.4) / \sqrt{(1.2^2 + 0.5^2)} = \mathbf{3.2\sigma}$ - vs ACT: $(71.5 - 67.9) / \sqrt{(1.2^2 + 1.5^2)} = \mathbf{1.9\sigma}$ - vs BOSS BAO: $(71.5 - 67.6) / \sqrt{(1.2^2 + 1.1^2)} = \mathbf{2.4\sigma}$

Agreement with late-universe measurements: - vs SH0ES: $(73.04 - 71.5) / \sqrt{(1.04^2 + 1.2^2)} = \mathbf{0.9\sigma}$ ✓ - vs TRGB: $(71.5 - 69.8) / \sqrt{(1.2^2 + 1.7^2)} = \mathbf{0.8\sigma}$ ✓ - vs GW170817: $(71.5 - 70.0) / \sqrt{(1.2^2 + 10^2)} = \mathbf{0.1\sigma}$ ✓ - vs Megamasers: $(73.9 - 71.5) / \sqrt{(3.0^2 + 1.2^2)} = \mathbf{0.7\sigma}$ ✓

1.6.3 5.3 Key Insight

The Zimmerman H_0 shows: - **Significant tension (2-3σ)** with all CMB-based measurements - **Good agreement (<1σ)** with most local measurements - **Best agreement** with GW170817 (standard siren, completely independent method)

This pattern suggests: 1. The true H_0 is likely ~70-72 km/s/Mpc 2. Planck may have a systematic bias of ~4 km/s/Mpc 3. SH0ES may have a smaller systematic bias of ~1-2 km/s/Mpc

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1.7 6. Why This Measurement Is Independent

1.7.1 6.1 Not the Distance Ladder

The SH0ES/Cepheid measurement requires: 1. Parallax distances to nearby Cepheids (Gaia) 2. Cepheid period-luminosity calibration 3. Cepheids in SNe Ia host galaxies 4. SNe Ia Hubble diagram

Our method requires none of these. We use: 1. Galaxy rotation curves (kinematic, not photometric) 2. Baryonic mass estimates (from stellar population models) 3. The Radial Acceleration Relation to determine a_0

1.7.2 6.2 Not CMB Physics

The Planck measurement requires: 1. CMB power spectrum measurement 2. Sound horizon calculation (r_s) 3. Assumed ΛCDM cosmology 4. Recombination physics at $z \approx 1100$

Our method requires none of these. We measure a_0 at $z \approx 0$ from local galaxies.

1.7.3 6.3 Shared Systematics?

Systematic	Affects Planck?	Affects SH0ES?	Affects Zimmerman?
Sound horizon	Yes	No	**No**
Cepheid calibration	No	Yes	**No**
SNe Ia standardization	No	Yes	**No**
CMB foregrounds	Yes	No	**No**
Parallax zero-point	No	Yes	**No**
Stellar M/L ratios	No	No	**Weak (~5%)**
Galaxy distances	No	Yes	**No** (cancels)
ΛCDM assumption	Yes	Weak	**No**

The Zimmerman method shares essentially **no systematic errors** with either the CMB or distance ladder approaches.

1.8 7. Addressing Objections

1.8.1 7.1 "a_0 is empirical, not fundamental"

Response: The Zimmerman Formula **derives** a_0 from the cosmological critical density:

$$\frac{\quad}{\quad}$$

This transforms a_0 from an empirical constant into a cosmologically-determined quantity. Whether or not MOND is the correct theory, the numerical relationship $a_0 = cH_0/5.79$ holds to 0.5% accuracy.

1.8.2 7.2 "This assumes MOND is correct"

Response: We do **not** assume MOND is the correct theory of gravity. We simply use the observed fact that: - Galaxy rotation curves show a characteristic acceleration scale a_0 - This scale is measured to be $(1.20 \pm 0.02) \times 10^{-10}$ m/s² - The Zimmerman Formula relates a_0 to H_0

Whether a_0 arises from modified gravity, dark matter properties, or some other physics is irrelevant to the H_0 measurement.

1.8.3 7.3 "The coefficient 5.79 is arbitrary"

Response: The coefficient is **both derived AND empirically verified**:

1. **Derived:** $5.79 = 2\sqrt{(8\pi/3)}$ emerges from the Friedmann equation structure
2. **Verified:** Using Planck's ϱ_c, the formula $a_0 = c\sqrt{(G\varrho_c)}/2$ predicts $a_0 = 1.19 \times 10^{-10}$ m/s²
3. **Observed:** Galaxy rotation curves give $a_0 = (1.20 \pm 0.02) \times 10^{-10}$ m/s²

The prediction matches observation within **<1%**. Any other coefficient would fail this test. This is not a free parameter — it is determined by nature.

1.8.4 7.4 "This is circular reasoning"

Response: The reasoning is explicitly **non-circular**:

1. **Independent measurement #1:** Galaxy dynamicists measured $a_0 = 1.2 \times 10^{-10}$ m/s² from rotation curves (no cosmology used)
2. **Independent measurement #2:** Cosmologists measured ϱ_c = 9.47×10^{-27} kg/m³ from the CMB (no galaxy dynamics used)
3. **The discovery:** These independently-measured quantities satisfy $a_0 = c\sqrt{(G\varrho_c)}/2$ within <1%

The circularity test: **Could these measurements have disagreed?** Yes — if a_0 were 10^{-9} or 10^{-11} m/s², the formula would fail. The fact that they agree is the non-trivial result.

This is no different from Cavendish measuring G in a laboratory, which then predicts planetary orbits — not circular, just physics.

1.8.5 7.5 "a_0 might vary between galaxies"

Response: The RAR shows intrinsic scatter of only 0.13 dex, implying a_0 is universal to ~30%. More careful analyses (McGaugh+ 2016) find no evidence for variation with galaxy type, mass, surface brightness, or environment. If a_0 varied significantly, the BTFR would show much larger scatter than observed.

1.8.6 7.6 "What about systematic errors in M★?"

Response: Stellar mass-to-light (M/L) ratios have ~0.1 dex uncertainty. However: 1. The RAR uses the **ratio** g_obs/g_bar, reducing sensitivity 2. The a_0 determination averages over 153+ galaxies 3. Net effect on a_0: <5%, or <4 km/s/Mpc in H_0

This is already included in our 1.2 km/s/Mpc uncertainty.

1.9 8. Implications for the Hubble Tension

1.9.1 8.1 Where Is the True H_0?

The Zimmerman measurement suggests $H_0 \approx 71.5$ km/s/Mpc, which implies: - **Planck is biased low by ~4 km/s/Mpc** (3.2σ) - **SH0ES is biased high by ~1.5 km/s/Mpc** (0.9σ)

1.9.2 8.2 Possible Resolutions

If $H_0 \approx 71.5$ is correct:

1. **Planck systematics:** The sound horizon r_s may be ~3% larger than assumed, perhaps due to:
   - Early dark energy
   - Additional radiation
   - Modified recombination physics
2. **SH0ES systematics:** The Cepheid calibration may be ~2% too high, perhaps due to:
   - Crowding/blending effects
   - Metallicity dependence
   - Period-luminosity relation calibration

1.9.3 8.3 Consilience with Other Methods

The methods closest to $H_0 = 71.5$: - **GW170817:** 70.0 (completely independent, no EM calibration) - **TRGB:** 69.8 (different stellar physics from Cepheids) - **Carnegie-Chicago:** 69.6 (independent calibration)

These methods share the characteristic of being **independent** of the standard distance ladder or CMB assumptions.

1.10 9. Predictions and Tests

1.10.1 9.1 Future a_0 Measurements

As a_0 measurements improve: - GAIA DR4 proper motions → $\delta a_0 \sim 1\%$ - Expanded SPARC-like samples → $\delta a_0 \sim 1\%$ - This would yield $\delta H_0 \sim 0.7$ km/s/Mpc

1.10.2 9.2 Consistency Checks

The Zimmerman Formula makes additional predictions: - **Critical density:** $\varrho_c = 9.6 \times 10^{-27}$ kg/m³ (1.4% match to Planck) - **Cosmological constant:** $\Lambda = 1.2 \times 10^{-52}$ m⁻² (13% match)

These provide independent consistency checks.

1.10.3 9.3 Falsification

The Zimmerman H_0 is falsified if: 1. **a_0 varies significantly between galaxies** (observed scatter is only 0.13 dex) 2. **Future H_0 measurements converge to 67 or 74** (not ~71) 3. **The coefficient 5.79 is wrong** (but it's mathematically derived)

1.11 10. Summary

We have presented a new, independent measurement of the Hubble constant:

This is derived from:

where $a_0 = (1.20 \pm 0.02) \times 10^{-10}$ m/s² is the MOND acceleration scale measured from galaxy rotation curves, and $5.79 = 2\sqrt{(8\pi/3)}$ is derived from the Friedmann equation.

Key findings: 1. **Independent method:** Uses galaxy kinematics, not distance ladder or CMB 2. **Between Planck and SH0ES:** 3.2σ tension with Planck, 0.9σ with SH0ES 3. **Best agreement with GW170817:** The only other purely kinematic method 4. **Suggests true $H_0 \approx 71$:** With systematics in both CMB and Cepheid methods

The Zimmerman method provides a **third way** to measure H_0 that may help resolve the Hubble tension.

1.12 Appendix A: Detailed Error Budget

Source	Contribution to σ_H_0
Statistical (a_0 measurement)	1.0 km/s/Mpc

Source	Contribution to σ_H_0
M/L systematic	0.5 km/s/Mpc
Interpolation function	0.3 km/s/Mpc
Sample selection	0.2 km/s/Mpc
Total (quadrature)	**1.2 km/s/Mpc**

1.13 Appendix B: Comparison to Previous Work

The relationship $a_0 \approx cH_0$ has been noted before (Milgrom 1983, Sanders 1990), but always as a "cosmic coincidence." The Zimmerman Formula is the first to: 1. **Derive** the exact coefficient (5.79) 2. **Invert** the relationship to measure H_0 3. **Provide rigorous error analysis** 4. **Compare to all H_0 measurements systematically**

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Reproducibility: All calculations can be verified with the code at:
https://github.com/carlzimmerman/zimmerman-formula/tree/main/paper/analysis

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